United States Patent Application Publication	20250287585
Kind Code	A1
Publication Date	September 11, 2025
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MANUFACTURING METHOD OF SEMICONDUCTOR STRUCTURE

Abstract

A manufacturing method of a semiconductor structure including the following steps is provided. A substrate is provided. The substrate includes a peripheral region and a memory region. A first isolation structure is formed in the substrate in the peripheral region. After the first isolation structure is formed, a first floating gate layer and a tunneling dielectric layer are formed in the memory region. The first floating gate layer is located on the substrate. The tunneling dielectric layer is located between the first floating gate layer and the substrate. After the first floating gate layer and the tunneling dielectric layer are formed, a second isolation structure is formed in the substrate in the memory region.

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Appl. No.: **18/613139**

Filed: **March 22, 2024**

Foreign Application Priority Data

TW	113107851	Mar. 05, 2024

Publication Classification

Int. Cl.: **H10B41/42** (20230101); **H01L21/311** (20060101); **H01L21/762** (20060101); **H01L29/66** (20060101)

U.S. Cl.:

CPC **H10B41/42** (20230201); **H01L21/31116** (20130101); **H01L21/31144** (20130101); **H01L21/76224** (20130101); **H10D30/0411** (20250101); **H10D64/015** (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application claims the priority benefit of Taiwan application serial no. 113107851, filed on Mar. 5, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.
BACKGROUND
Technical Field
[0002] The invention relates to a manufacturing method of a semiconductor structure, and particularly relates to a manufacturing method of a semiconductor structure including a floating gate layer and a tunneling dielectric layer.
Description of Related Art
[0003] Since the non-volatile memory has the advantage that the stored data will not disappear even after being powered off, many electronic products must require this type of memory to maintain normal operation when the electronic products are turned on. However, how to improve the reliability and the operation speed of the memory device is the goal of continuous efforts.
SUMMARY
[0004] The invention provides a manufacturing method of a semiconductor structure, which can improve the reliability and the operation speed of the memory device.
[0005] The invention provides a manufacturing method of a semiconductor structure, which includes the following steps. A substrate is provided. The substrate includes a peripheral region and a memory region. A first isolation structure is formed in the substrate in the peripheral region. After the first isolation structure is formed, a first floating gate layer and a tunneling dielectric layer are formed in the memory region. The first floating gate layer is located on the substrate. The tunneling dielectric layer is located between the first floating gate layer and the substrate. After the first floating gate layer and the tunneling dielectric layer are formed, a second isolation structure is formed in the substrate in the memory region.
[0006] Based on the above description, in the manufacturing method of the semiconductor structure according to the invention, before the second isolation structure is formed in the substrate in the memory region, the first floating gate layer and the tunneling dielectric layer are formed in the memory region. Therefore, the corner thinning of the tunneling dielectric layer can be prevented, so that the tunneling dielectric layer can have a uniform thickness, thereby improving the reliability of the memory device. In addition, the thickness of the tunneling dielectric layer can be adjusted according to requirements, thereby improving the operation speed of the memory device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1**A to FIG. **1**U are cross-sectional views of a manufacturing process of a semiconductor structure according to some embodiments of the invention.
DESCRIPTION OF THE EMBODIMENTS
[0008] Referring to FIG. **1**A, a substrate **100** is provided. The substrate **100** includes a peripheral region R**1** and a memory region R**2**. The peripheral region R**1** may be a peripheral circuit region.

The peripheral region R**1** may include a first region R**11** and a second region R**12**. In some embodiments, the first region R**11** may be a high-voltage device region (e.g., high-voltage transistor region), and the second region R**12** may be a low-voltage device region (e.g., low-voltage transistor region). The memory region R**2** may be a flash memory region such as a NOR flash memory region. The substrate **100** may be a semiconductor substrate such as a silicon substrate.

[0009] A pad layer **102** may be formed on the substrate **100**. The material of the pad layer **102** may include silicon oxide. The method of forming the pad layer **102** may include a thermal oxidation method. Required doped regions (e.g., well region) (not shown) may be formed in the substrate **100** in the memory region R**2**.

[0010] Referring to FIG. **1**B, the pad layer **102** in memory region R**2** may be removed. In some embodiments, the pad layer **102** in the memory region R**2** may be removed by a lithography process and an etching process.

[0011] Referring to FIG. **1**C, a dielectric material layer **104**, a floating gate material layer **106**, and a hard mask layer **108** may be sequentially formed on the substrate **100**. The dielectric material layer **104** in the peripheral region R**1** may be located on the pad layer **102**. The material of the dielectric material layer **104** may include silicon oxide. The method of forming the dielectric material layer **104** may include a thermal oxidation method. The material of the floating gate material layer **106** may include undoped polysilicon or doped polysilicon. The method of forming the floating gate material layer **106** may include a chemical vapor deposition (CVD) method. The material of the hard mask layer **108** may include silicon nitride. The method of forming the hard mask layer **108** may include a CVD method.

[0012] Referring to FIG. **1**D, the hard mask layer **108** and the floating gate material layer **106** in the peripheral region R**1** are patterned to form a patterned hard mask layer **108***a* and a patterned floating gate material layer **106***a* in the peripheral region R**1**. In some embodiments, the hard mask layer **108** and the floating gate material layer **106** in the peripheral region R**1** may be patterned by a lithography process and an etching process.

[0013] A spacer material layer **110** may be conformally formed on the patterned hard mask layer **108***a*, the patterned floating gate material layer **106***a*, the dielectric material layer **104**, and the hard mask layer **108**. The method of forming the spacer material layer **110** may include an atomic layer deposition method.

[0014] Referring to FIG. **1**E, the spacer material layer **110**, the dielectric material layer **104**, the pad layer **102**, and the substrate **100** may be patterned to form a spacer **110***a* and to form a trench T**1** in the substrate **100**. The spacer **110***a* is located on the sidewall and the top surface of the patterned hard mask layer **108***a* and is located on the sidewall of the patterned floating gate material layer **106***a*. The spacer **110***a* may be further located on the dielectric material layer **104**. In some embodiments, the spacer material layer **110**, the dielectric material layer **104**, the pad layer **102**, and the substrate **100** may be patterned by a lithography process and an etching process.

[0015] Referring to FIG. **1**F, the spacer **110***a* may be removed. In the process of removing the spacer **110***a*, a portion of the dielectric material layer **104** and a portion of the pad layer **102** located directly below the spacer **110***a* may be simultaneously removed. The method of removing the spacer **110***a* may include a wet etching method.

[0016] Referring to FIG. **1**G, an isolation structure **112** may be formed in the trench T**1**. Therefore, the isolation structure **112** may be formed in the substrate **100** in the peripheral region R**1**. The isolation structure **112** may be located on the sidewall of the patterned hard mask layer **108***a* and the sidewall of the patterned floating gate material layer **106***a*. The isolation structure **112** may be a single-layer structure or a multilayer structure. In the present embodiment, the isolation structure **112** is, for example, a multilayer structure. For example, the isolation structure **112** may include a liner layer **114**, a dielectric layer **116**, and a dielectric layer **118**. The liner layer **114**, the dielectric layer **116**, and the dielectric layer **118** are sequentially located in the trench T**1**. The material of the liner layer **114** may include silicon oxide formed by a CVD process. The material of the dielectric layer **116** may include silicon oxide formed by a high aspect ratio process (HARP). The material of the dielectric layer **118** may include silicon oxide formed by a CVD process. In some embodiments, the method of forming the liner layer **114**, the dielectric layer **116**, and the dielectric layer **118** may include the following steps. First, the material layer of the liner layer **114**, the material layer of the dielectric layer **116**, and the material layer of the dielectric layer **118** filling the trench T**1** may be sequentially formed. Then, a chemical mechanical polishing (CMP) process may be performed on the material layer of the dielectric layer **118**, the material layer of the dielectric layer **116**, and the material layer of the liner layer **114** to form the dielectric layer **118**, the dielectric layer **116**, and the liner layer **114**.

[0017] Referring to FIG. **1**H, the hard mask layer **108**, the floating gate material layer **106**, and the dielectric material layer **104** in the memory region R**2** may be patterned to form a patterned hard mask layer **108***b*, a floating gate layer **106***b*, and a tunneling dielectric layer **104***a*. Therefore, after the isolation structure **112** is formed, the floating gate layer **106***b* and the tunneling dielectric layer **104***a* may be formed in the memory region R**2**. The floating gate layer **106***b* is located on the substrate **100**. The tunneling dielectric layer **104***a* is located between the floating gate layer **106***b* and the substrate **100**. In addition, the substrate **100** in the memory region R**2** may be patterned to form a trench T**2** in the substrate **100**. In some embodiments, the hard mask layer **108**, the floating gate material layer **106**, the dielectric material layer **104**, and the substrate **100** in the memory region R**2** may be patterned by a lithography process and an etching process.

[0018] Referring to FIG. **1**I, an isolation structure **120** may be formed in the trench T**2**. Therefore, after the floating gate layer **106***b* and the tunneling dielectric layer **104***a* are formed, the isolation structure **120** may be formed in the substrate **100** in the memory region R**2**. The depth D**1** of the isolation structure **112** located in the substrate **100** may be greater than the depth D**2** of the isolation structure **120** located in the substrate **100**. The width W**1** of the isolation structure **112** located in the substrate **100** may be greater than the width W**2** of the isolation structure **120** located in the substrate **100**. The isolation structure **120** may be a single-layer structure or a multilayer structure. In the present embodiment, the isolation structure **120** is, for example, a multilayer structure. For example, the isolation structure **120** may include a liner layer **122** and a dielectric layer **124**. The liner layer **122** and the dielectric layer **124** are sequentially located the in the trench T**2**. The material of the liner layer **122** may include silicon oxide formed by a CVD process. The material of the dielectric layer **124** may include silicon oxide formed by a high aspect ratio process (HARP). In some embodiments, a method of forming the liner layer **122** and the dielectric layer **124** may include the following steps. First, the material layer of the liner layer **122** and the material layer of the dielectric layer **124** filling the trench T**2** may be sequentially formed. Then, a CMP process may be performed on the material layer of the dielectric layer **124** and the material layer of the liner layer **122** to form the dielectric layer **124** and the liner layer **122**.

[0019] Referring to FIG. **1**J, a patterned photoresist layer **126** may be formed in the peripheral region R**1**. The patterned photoresist layer **126** may cover the isolation structure **112**. The patterned photoresist layer **126** may further cover the patterned hard mask layer **108***a*. The patterned photoresist layer **126** may be formed by a lithography process.

[0020] An etch back process (e.g., dry etching process) may be performed on the isolation structure **120** by using the patterned photoresist layer **126** as a mask to reduce the height of the isolation structure **120**. The isolation structure **120** may be located on the sidewall of the tunneling dielectric layer **104***a* and the sidewall of the floating gate layer **106***b*. The top surface S**1** of the isolation structure **120** may be higher than the bottom surface S**2** of the floating gate layer **106***b* and lower than the top surface S**3** of the floating gate layer **106***b*.

[0021] Referring to FIG. **1**K, the patterned photoresist layer **126** may be removed. The method of removing the patterned photoresist layer **126** may include a dry stripping method or a wet stripping method.

[0022] The patterned hard mask layers **108***a* and **108***b* may be removed. The method of removing the patterned hard mask layers **108***a* and **108***b* may include a wet etching method.

[0023] Referring to FIG. **1**L, a floating gate material layer **128** may be formed on the floating gate layer **106***b*. The floating gate material layer **128** may be further formed on the isolation structure **120**, the isolation structure **112**, and the patterned floating gate material layer **106***a*. The material of the floating gate material layer **128** may include undoped polysilicon or doped polysilicon. The method of forming the floating gate material layer **128** may include a CVD method.

[0024] A hard mask layer **130**, a hard mask layer **132**, and a hard mask layer **134** may be sequentially formed on the floating gate material layer **128**. The materials of the hard mask layers **130** and **134** may include silicon oxide. The material of the hard mask layer **132** may include silicon nitride. The method of forming hard mask layers **130**, **132**, and **134** may include a CVD method.

[0025] Referring to FIG. **1**M, a portion of the hard mask layer **134**, a portion of the hard mask layer **132**, a portion of the hard mask layer **130**, a portion of the floating gate material layer **128**, and the patterned floating gate material layer **106***a* in the peripheral region R**1** may be removed. In some embodiments, a portion of the hard mask layer **134**, a portion of the hard mask layer **132**, a portion of the hard mask layer **130**, a portion of the floating gate material layer **128**, and the patterned floating gate material layer **106***a* in the peripheral region R**1** may be removed by a lithography process and an etching process. In the above etching process, a portion of the isolation structure **112** may be removed. Required doped regions (e.g., well region) (not shown) may be formed in the substrate **100** in the peripheral region R**1**.

[0026] Referring to FIG. **1**N, the dielectric material layer **104** and the pad layer **102** in the peripheral region R**1** may be removed. The method of removing the dielectric material layer **104** and the pad layer **102** in the peripheral region R**1** may include a wet etching method. In the process of removing the dielectric material layer **104** and the pad layer **102** in the peripheral region R**1**, the hard mask layer **134**, a portion of the hard mask layer **130**, and a portion of the isolation structure **112** may be removed.

[0027] A gate dielectric layer **136** and a gate dielectric layer **138** may be formed on the substrate **100** in the peripheral region R**1**. The gate dielectric layer **136** and the gate dielectric layer **138** may be respectively located in the first region R**11** and the second region R**12**. In the process of forming the gate dielectric layer **136** and the gate dielectric layer **138**, a dielectric layer **140** may be simultaneously formed on the floating gate material layer **128**. The materials of the gate dielectric layer **136**, the gate dielectric layer **138**, and the dielectric layer **140** may include silicon oxide. The method of forming the gate dielectric layer **136**, the gate dielectric layer **138**, and the dielectric layer **140** may include a thermal oxidation method.

[0028] A patterned photoresist layer **142** may be formed in the memory region R**2** and the first region R**11**. The patterned photoresist layer **142** may expose the gate dielectric layer **138** and a portion of the isolation structure **112** in the second region R**12**. The patterned photoresist layer **142** may be formed by a lithography process.

[0029] Referring to FIG. **1**O, the dielectric layer **138** may be removed by using the patterned photoresist layer **142** as a mask. In the process of removing the gate dielectric layer **138**, a portion of the isolation structure **112** exposed by the patterned photoresist layer **142** may be removed. In some embodiments, the method of removing the gate dielectric layer **138** and a portion of the isolation structure **112** may include a wet etching method.

[0030] Referring to FIG. **1**P, the patterned photoresist layer **142** may be removed. The method of removing the patterned photoresist layer **142** may include a dry stripping method or a wet stripping method.

[0031] A gate dielectric layer **144** may be formed on the substrate **100** in the peripheral region R**1** (e.g., second region R**12**). The gate dielectric layer **136** and the gate dielectric layer **144** may be separated from each other. The thickness TK**1** of the gate dielectric layer **136** may be greater than the thickness TK**2** of the gate dielectric layer **144**. The method of forming the gate dielectric layer

144 may include a thermal oxidation method.
[0032] Referring to FIG. **1**Q, a gate material layer **146** may be formed on the gate dielectric layer **136** and the gate dielectric layer **144**. The gate material layer **146** may be further formed on the hard mask layer **132** in the memory region R**2**. The material of gate material layer **146** may include doped polysilicon. The method of forming the gate material layer **146** may include a CVD method.
[0033] Referring to FIG. **1**R, the gate material layer **146** in the memory region R**2** may be removed to expose the hard mask layer **132**. In some embodiments, the gate material layer **146** in the memory region R**2** may be removed by a lithography process and an etching process.
[0034] Referring to FIG. **1**S, the hard mask layer **132** in the memory region R**2** may be removed. The method of removing the hard mask layer **132** in the memory region R**2** may include a wet etching method. The hard mask layer **130** in memory region R**2** may be removed. The method of removing the hard mask layer **130** in the memory region R**2** may include a wet etching method. In some embodiments, a portion of the hard mask layer **132** and a portion of the hard mask layer **130** may remain directly above the isolation structure **112** located at the edge of the first region R**11**.
[0035] Referring to FIG. **1**T, the floating gate material layer **128** may be patterned to form a floating gate layer **128***a* and an opening OP**1**. Therefore, the floating gate layer **128***a* may be formed on the floating gate layer **106***b*. A portion of the floating gate layer **128***a* may be located on the isolation structure **120**. The width W**3** of the floating gate layer **128***a* may be greater than the width W**4** of the floating gate layer **106***b*. In some embodiments, the floating gate material layer **128** may be patterned by a lithography process and an etching process. In some embodiments, the etching process may be a dry etching process, thereby forming the opening OP**1** with the required specification. In this way, it helps to improve the process window of the subsequent process of forming a control gate **148** (FIG. **1**U) in the opening OP**1**.
[0036] Referring to FIG. **1**U, a control gate **148** may be formed on the floating gate layer **128***a* and in the opening OP**1**. The material of the control gate **148** may include doped polysilicon. In addition, a dielectric layer **150** may be formed between the control gate **148** and the floating gate layer **128***a*. The dielectric layer **150** may be a single-layer structure or a multilayer structure. In some embodiments, the dielectric layer **150** may be an oxide/nitride/oxide (ONO) composite layer.
[0037] In subsequent processes, the gate material layer **146** may be patterned to respectively form a gate for a high-voltage device (e.g., high-voltage transistor device) and a gate for a low-voltage device (e.g., low-voltage transistor device) in the first region R**1** and the second region R**2**, respectively, and the description thereof is omitted here.
[0038] Based on the above embodiments, in the manufacturing method of the semiconductor structure **10**, before the isolation structure **120** is formed in the substrate **100** in the memory region R**2**, the floating gate layer **106***b* and the tunneling dielectric layer **104***a* are formed in the memory region R**2**. Therefore, the corner thinning of the tunneling dielectric layer **104***a* can be prevented, so that the tunneling dielectric layer **104***a* can have a uniform thickness, thereby improving the reliability of the memory device (e.g., NOR flash memory). In addition, the thickness of the tunneling dielectric layer **104***a* can be adjusted according to requirements, thereby improving the operation speed of the memory device (e.g., NOR flash memory device).
[0039] Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the invention. Accordingly, the scope of the invention is defined by the attached claims not by the above detailed descriptions.

Claims

1. A manufacturing method of a semiconductor structure, comprising: providing a substrate, wherein the substrate comprises a peripheral region and a memory region; forming a first isolation structure in the substrate in the peripheral region; after forming the first isolation structure, forming a first floating gate layer and a tunneling dielectric layer in the memory region, wherein the first floating gate layer is located on the substrate, and the tunneling dielectric layer is located between the first floating gate layer and the substrate; and after forming the first floating gate layer and the tunneling dielectric layer, forming a second isolation structure in the substrate in the memory region.

2. The manufacturing method of the semiconductor structure according to claim 1, wherein a method of forming the first isolation structure comprises: sequentially forming a dielectric material layer, a floating gate material layer, and a hard mask layer on the substrate; patterning the hard mask layer and the floating gate material layer in the peripheral region to form a patterned hard mask layer and a patterned floating gate material layer in the peripheral region; conformally forming a spacer material layer on the patterned hard mask layer, the patterned floating gate material layer, and the dielectric material layer; patterning the spacer material layer, the dielectric material layer, and the substrate to form a spacer and to form a trench in the substrate, wherein the spacer is located on a sidewall and a top surface of the patterned hard mask layer and is located on a sidewall of the patterned floating gate material layer; removing the spacer; and forming the first isolation structure in the trench.

3. The manufacturing method of the semiconductor structure according to claim 2, further comprising: in the process of removing the spacer, simultaneously removing a portion of the dielectric material layer located directly below the spacer.

4. The manufacturing method of the semiconductor structure according to claim 2, wherein the first isolation structure is located on the sidewall of the patterned hard mask layer and the sidewall of the patterned floating gate material layer.

5. The manufacturing method of the semiconductor structure according to claim 1, wherein a method of forming the first floating gate layer and the tunneling dielectric layer comprises: sequentially forming a dielectric material layer, a floating gate material layer, and a hard mask layer on the substrate; and patterning the hard mask layer, the floating gate material layer, and the dielectric material layer in the memory region to form a patterned hard mask layer, the first floating gate layer, and the tunneling dielectric layer.

6. The manufacturing method of the semiconductor structure according to claim 5, wherein a method of forming the second isolation structure comprises: patterning the substrate in the memory region to form a trench in the substrate; and forming the second isolation structure in the trench.

7. The manufacturing method of the semiconductor structure according to claim 5, wherein the second isolation structure is located on a sidewall of the tunneling dielectric layer and a sidewall of the first floating gate layer.

8. The manufacturing method of the semiconductor structure according to claim 5, further comprising: forming a patterned photoresist layer is in the peripheral region, wherein the patterned photoresist layer covers the first isolation structure; performing an etch back process on the second isolation structure by using the patterned photoresist layer as a mask to reduce a height of the second isolation structure; removing the patterned photoresist layer; and removing the patterned hard mask layer.

9. The manufacturing method of the semiconductor structure according to claim 8, wherein a top surface of the second isolation structure is higher than a bottom surface of the first floating gate layer and lower than a top surface of the first floating gate layer.

10. The manufacturing method of the semiconductor structure according to claim 1, further comprising: forming a second floating gate layer on the first floating gate layer.

11. The manufacturing method of the semiconductor structure according to claim 10, wherein a width of the second floating gate layer is greater than a width of the first floating gate layer.

12. The manufacturing method of the semiconductor structure according to claim 10, wherein a portion of the second floating gate layer is located on the second isolation structure.

13. The manufacturing method of the semiconductor structure according to claim 10, wherein a method of forming the second floating gate layer comprises: forming a floating gate material layer on the first floating gate layer; and patterning the floating gate material layer to form the second floating gate layer and an opening.

14. The manufacturing method of the semiconductor structure according to claim 13, wherein the floating gate material layer is patterned by a lithography process and an etching process.

15. The manufacturing method of the semiconductor structure according to claim 14, wherein the etching process comprises a dry etching process.

16. The manufacturing method of the semiconductor structure according to claim 13, further comprising: forming a control gate on the second floating gate layer and in the opening; and forming a dielectric layer between the control gate and the second floating gate layer.

17. The manufacturing method of the semiconductor structure according to claim 1, further comprising: forming a first gate dielectric layer on the substrate in the peripheral region; forming a second gate dielectric layer on the substrate in the peripheral region, wherein the first gate dielectric layer and the second gate dielectric layer are separated from each other, and a thickness of the first gate dielectric layer is greater than a thickness of the second gate dielectric layer.

18. The manufacturing method of the semiconductor structure according to claim 17, further comprising: forming a gate material layer on the first gate dielectric layer and the second gate dielectric layer.

19. The manufacturing method of the semiconductor structure according to claim 1, wherein a depth of the first isolation structure located in the substrate is greater than a depth of the second isolation structure located in the substrate.

20. The manufacturing method of the semiconductor structure according to claim 1, wherein a width of the first isolation structure located in the substrate is greater than a width of the second isolation structure located in the substrate.